



**DEPARTMENT OF INFORMATION TECHNOLOGY
ENGINEERING LAB MANUAL**

CS23432 - SoftwareConstruction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

Name: V.CHARAN SUNDAR

Register No: 241001036

Year / Branch / Section: II/B.Tech/IT - A

Semester: IV

Academic Year: 2025-2026

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EXP NO: 1

AZUREDEVOPS ENVIRONMENT SETUP

Aim:

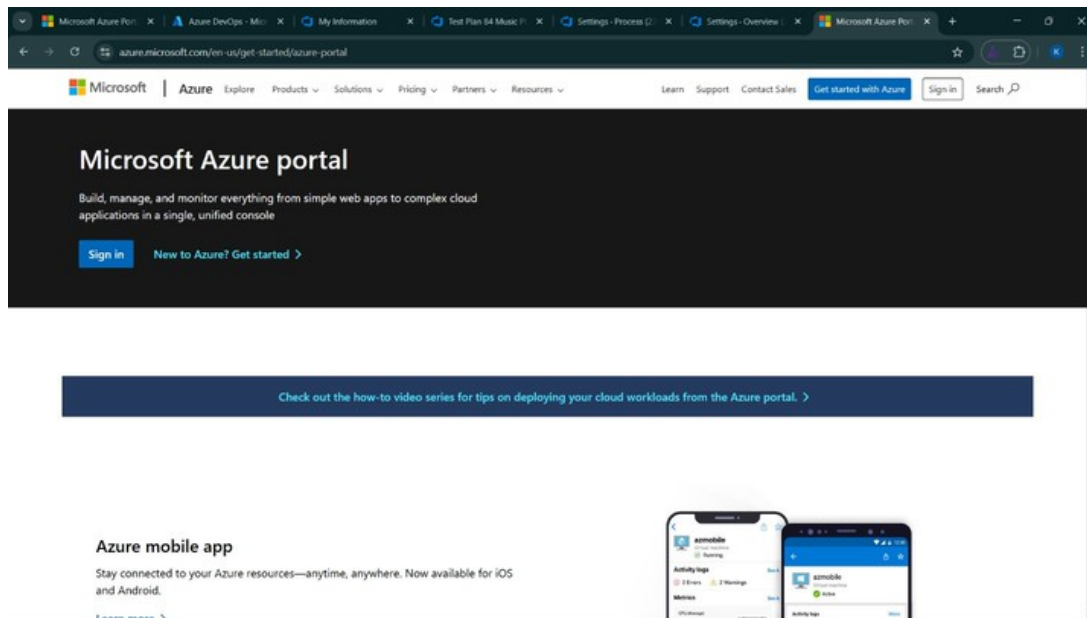
To set up and access the Azure DevOps environment by creating an organization through the Azure portal for the Hospital Management System project.

INSTALLATION

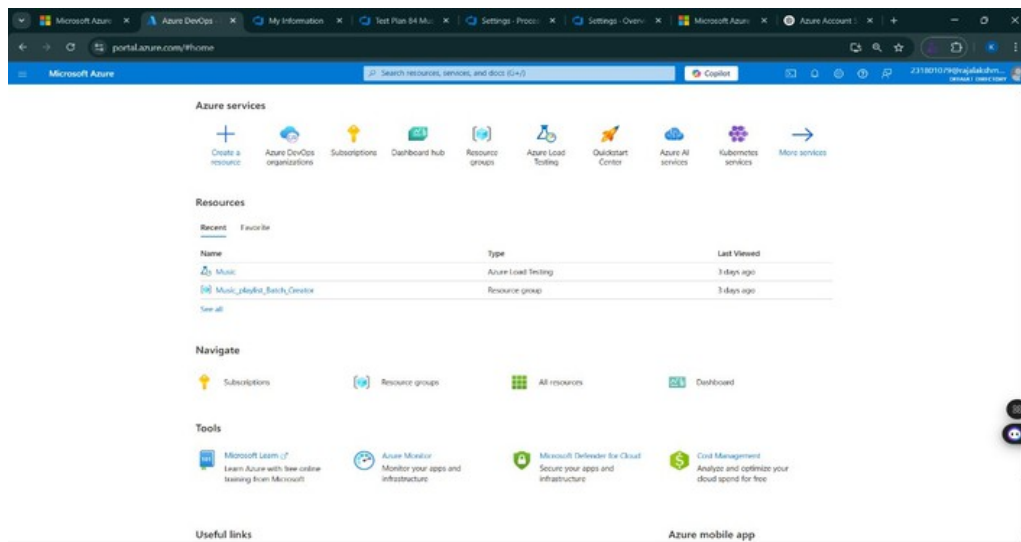
1. Open your web browser and go to the Azure website:

<https://azure.microsoft.com/en-us/get-started/azure-portal> . Sign in using your Microsoft account credentials.

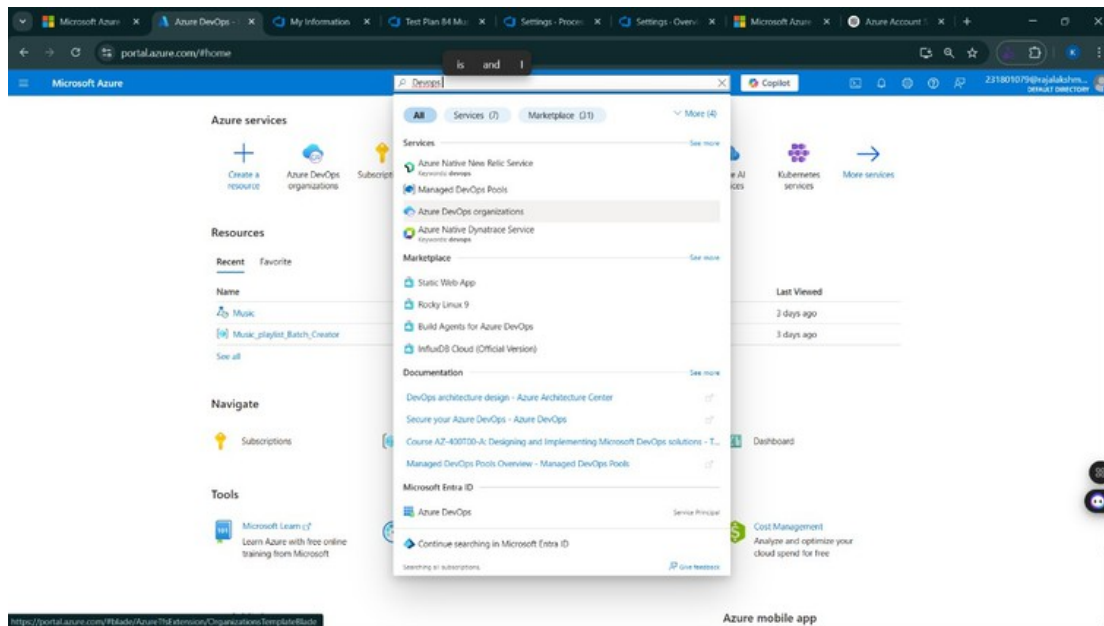
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



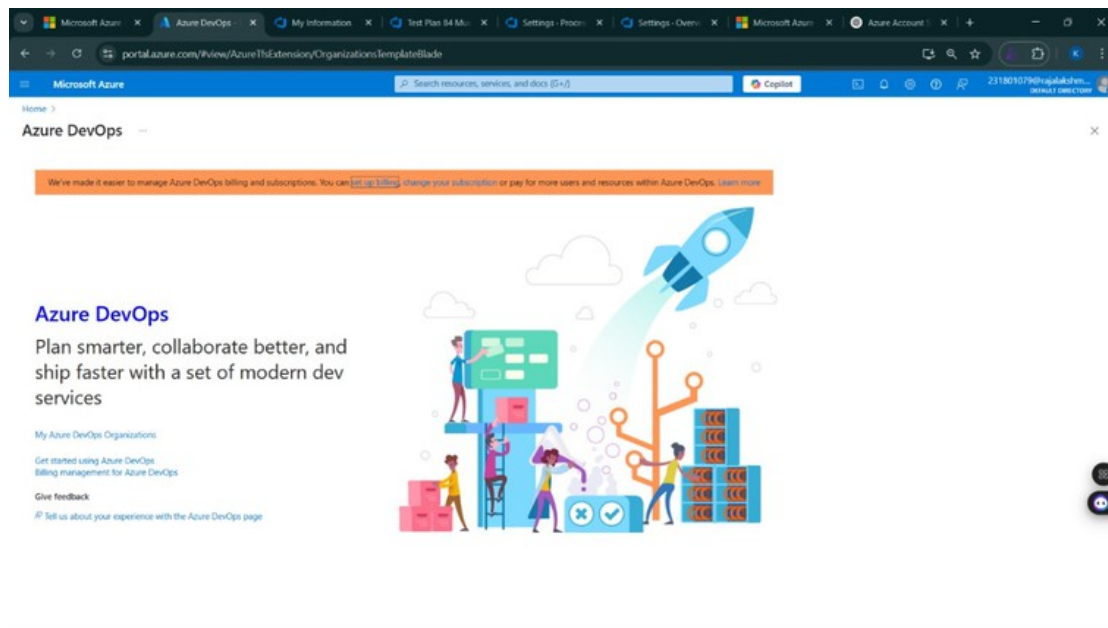
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

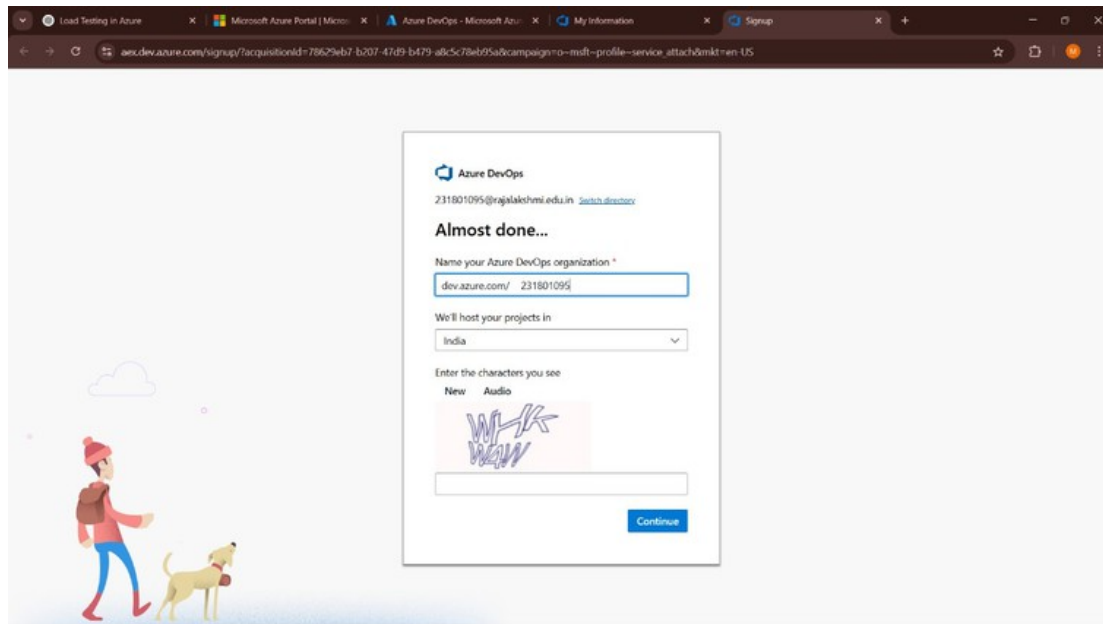
EXP NO: 2

AZUREDEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **Hospital Management System (HMS)**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project

×

Project name *

Hospital Management

Description

Visibility

Public

Anyone on the internet can view the project. Certain features like TFVC are not supported.

Private

Only people you give access to will be able to view this project.

By creating this project, you agree to the Azure DevOps [code of conduct](#)

^ Advanced

Version control ⓘ

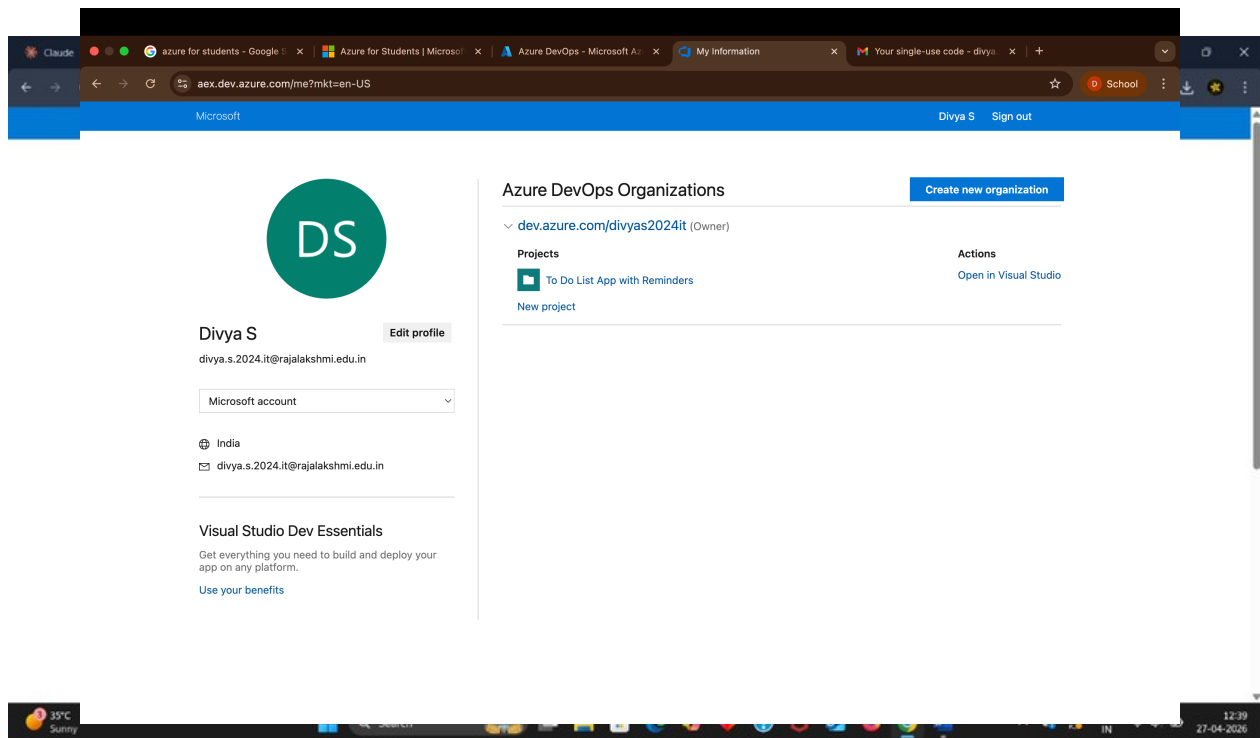
Git

Work item process ⓘ

231801095 Agile

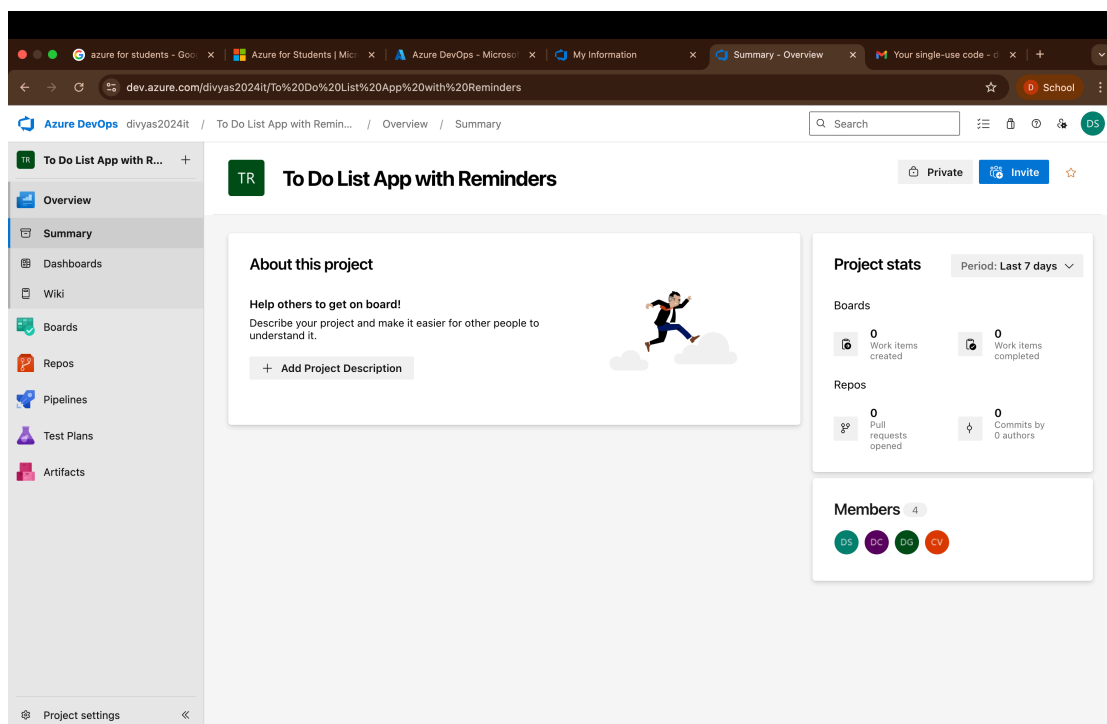
Cancel>Create

3. Once logged in, ensure you are in the correct organization. If you're part of multiple



organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

4. Project dashboard



5.To manage user stories:

a.From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b.On the **work items** page, you'll see the option to **Add a work item** at the top.
Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot shows the Azure DevOps interface for a project named 'To Do List App with Reminders'. The left-hand navigation menu is open, showing options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The 'Work items' section is selected, displaying a list of user stories. The table has columns for ID, Title, Assigned To, State, Area Path, and Tags. The first item is a user story assigned to Dinesh G, with state 'Active'. Other items are assigned to 'Unassigned' and have state 'New'.

ID	Title	Assigned To	State	Area Path	Tags
12	As a user, I want to add a new task so that I can keep track of...	Dinesh G	Active	To Do List App with Reminders	
13	As a user, I want to edit or delete tasks so that I can update my task	Unassigned	New	To Do List App with Reminders	
19	As a user, I want my tasks to be saved even after refresh so that I d	Unassigned	New	To Do List App with Reminders	
14	As a user, I want to mark tasks as completed so that I can track pro	Unassigned	New	To Do List App with Reminders	
16	As a user, I want to receive a notification when the reminder time is	Unassigned	New	To Do List App with Reminders	
15	As a user, I want to set a reminder date and time for a task so that I	Unassigned	New	To Do List App with Reminders	
17	As a user, I want to edit or delete reminders so that I can change m	Unassigned	New	To Do List App with Reminders	
11	As a user, I want to register and log in so that my tasks are sav...	Deepak Harinath C	New	To Do List App with Reminders	
18	As a user, I want a clean and simple UI so that the app is easy to us	Unassigned	New	To Do List App with Reminders	
51	Test data persistence	Unassigned	New	To Do List App with Reminders	
50	Load saved tasks on app startup	Unassigned	New	To Do List App with Reminders	
49	Implement localStorage / database storage	Unassigned	New	To Do List App with Reminders	
48	Optimize mobile view	Unassigned	New	To Do List App with Reminders	
47	Improve button & input styling	Unassigned	New	To Do List App with Reminders	
46	Design responsive layout	Unassigned	New	To Do List App with Reminders	

A screenshot of the Azure DevOps web interface showing a user story in the 'To Do List App with Reminders' project. The user story is titled '13 As a user, I want to edit or delete tasks so that I can update my task list.' and is in the 'New' state. The interface includes a sidebar with navigation options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main content area displays the user story details, including a description, acceptance criteria, planning information (Story Points, Priority, Risk), classification (Value area, Business), and a deployment section. The deployment section shows a link to an Azure Repos commit, pull request, or branch to see the status of your development. The related work section lists tasks such as 'Implement delete task option', 'Implement edit task functionality', and 'Test task updates'.

RESULT

Successfully created an Azure DevOps project with user story management and agile workflow

EX NO.3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

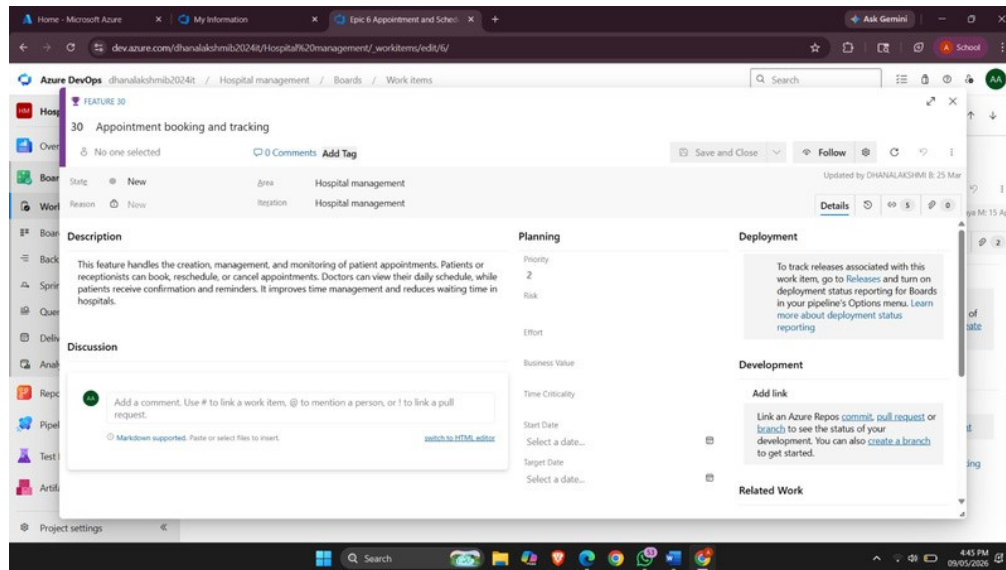
Create Epic, Features, User Stories, Task

1.Fill in Epics

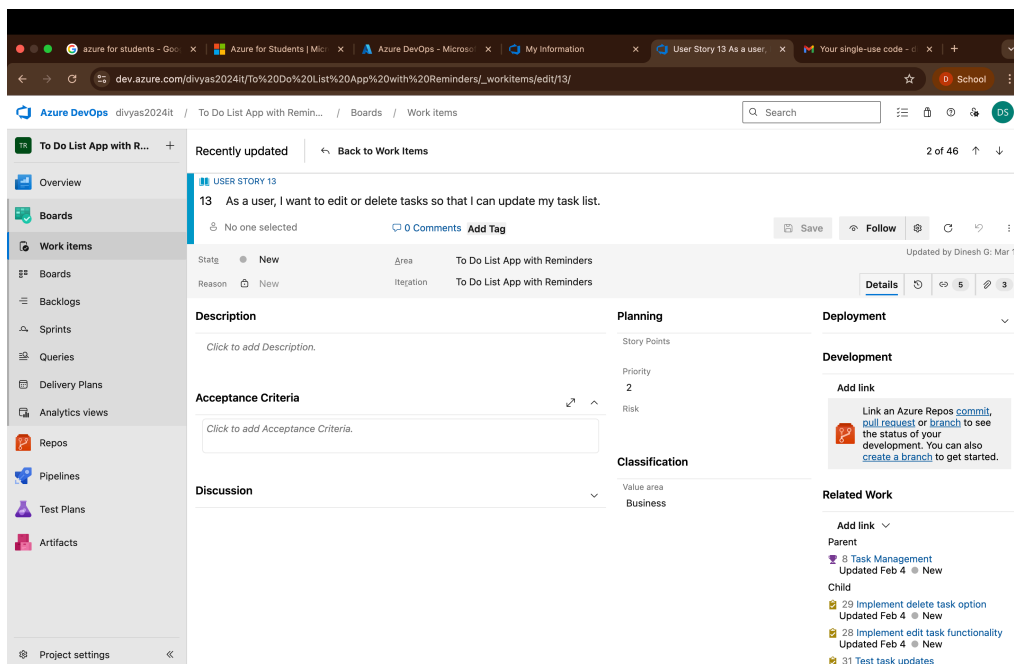
The screenshot displays the Azure DevOps Work Items page for a project named 'To Do List App with Reminders'. The left sidebar shows the project navigation menu with options like Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main area shows a list of work items under the 'Work items' tab. The list is filtered by 'Recently updated' and includes columns for ID, Title, Assigned To, State, Area Path, and Tags. The first item (ID 12) is assigned to Dinesh G and is in the 'Active' state. Other items are assigned to Deepak Harinath C or are 'Unassigned'.

ID	Title	Assigned To	State	Area Path	Tags
12	As a user, I want to add a new task so that I can keep track of...	Dinesh G	Active	To Do List App with Reminders	
13	As a user, I want to edit or delete tasks so that I can update my tas...	Unassigned	New	To Do List App with Reminders	
19	As a user, I want my tasks to be saved even after refresh so that I d...	Unassigned	New	To Do List App with Reminders	
14	As a user, I want to mark tasks as completed so that I can track prc...	Unassigned	New	To Do List App with Reminders	
16	As a user, I want to receive a notification when the reminder time is...	Unassigned	New	To Do List App with Reminders	
15	As a user, I want to set a reminder date and time for a task so that I...	Unassigned	New	To Do List App with Reminders	
17	As a user, I want to edit or delete reminders so that I can change m...	Unassigned	New	To Do List App with Reminders	
11	As a user, I want to register and log in so that my tasks are sav...	Deepak Harinath C	New	To Do List App with Reminders	
18	As a user, I want a clean and simple UI so that the app is easy to us...	Unassigned	New	To Do List App with Reminders	
51	Test data persistence	Unassigned	New	To Do List App with Reminders	
50	Load saved tasks on app startup	Unassigned	New	To Do List App with Reminders	
49	Implement localStorage / database storage	Unassigned	New	To Do List App with Reminders	
48	Optimize mobile view	Unassigned	New	To Do List App with Reminders	
47	Improve button & input styling	Unassigned	New	To Do List App with Reminders	
46	Design responsive layout	Unassigned	New	To Do List App with Reminders	

2.Fill in Features



3.Fill in User Story Details



Result:

Thus, the creation of epics, features, user story and task has been created successfully.

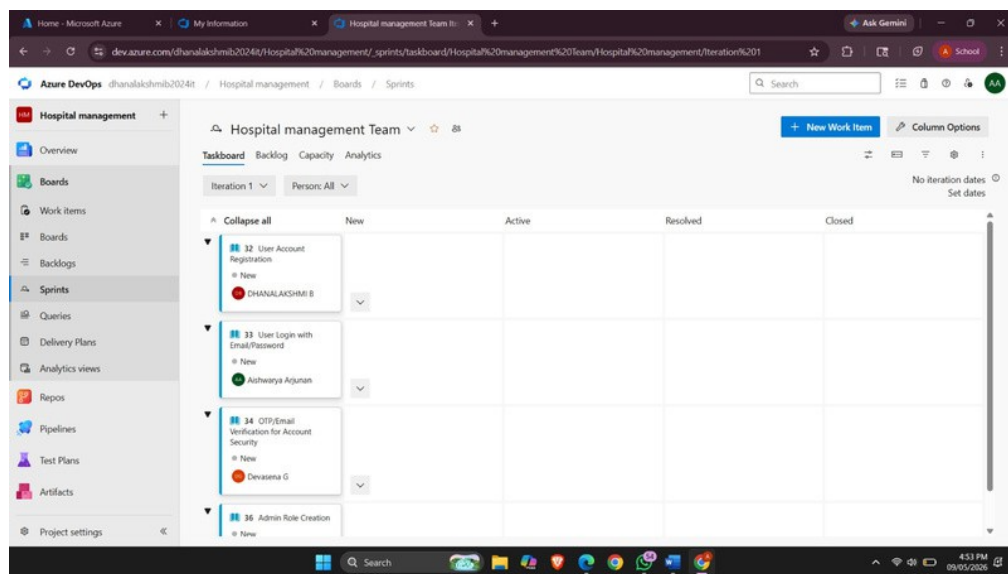
EXP NO: 4

SPRINT PLANNING

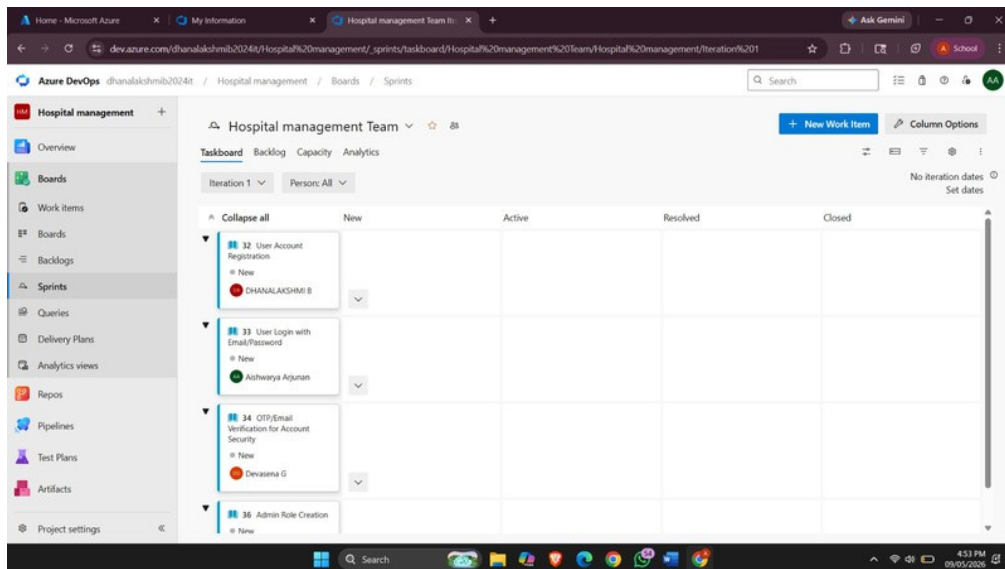
Aim:

To assign user stories to specific sprints for the Todo list app to setup reminders project.

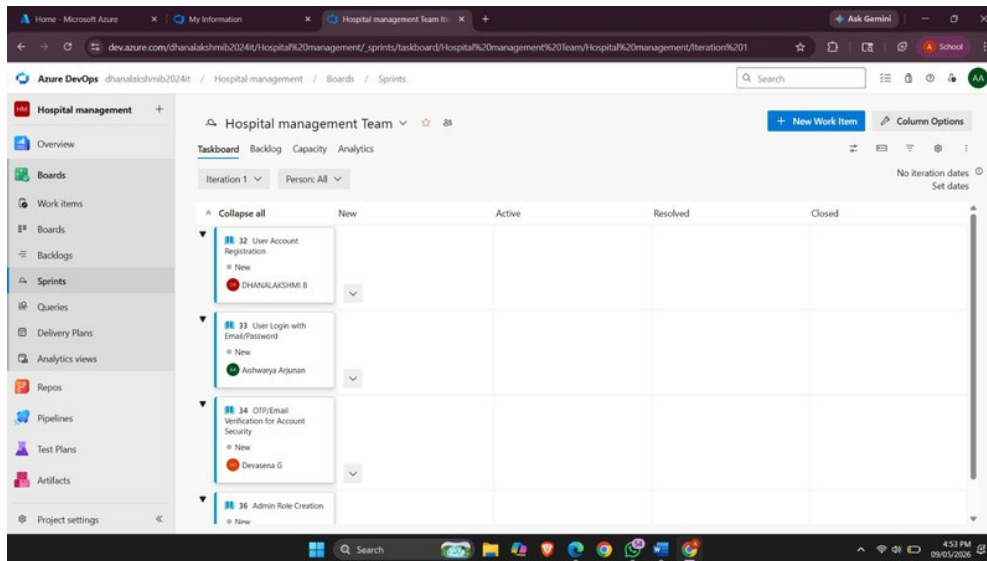
Sprint Planning Sprint 1



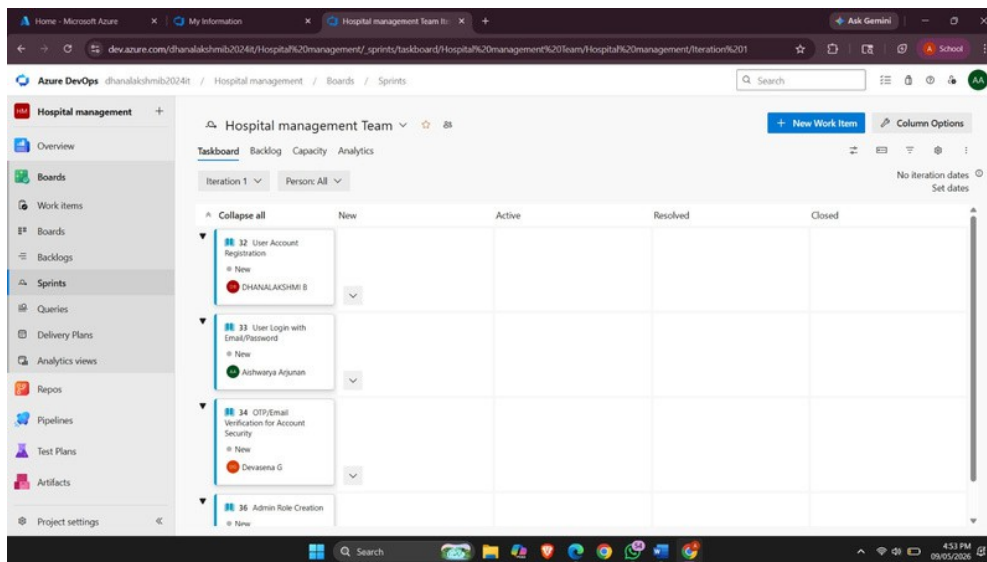
Sprint 2



Sprint 3



Sprint 4



RESULT

The sprints are created for the todo list app project

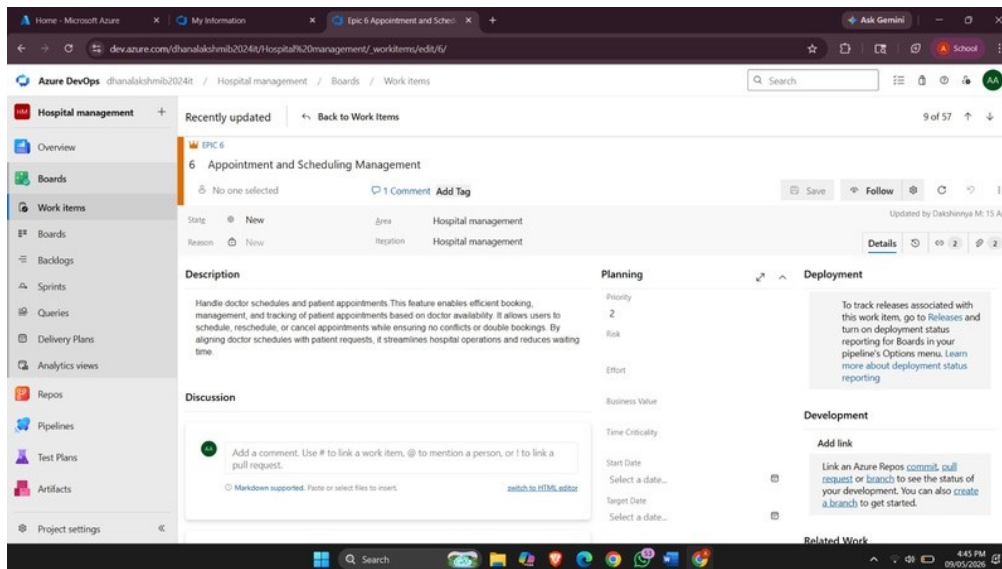
EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories of the Hospital Management System project.

Poker Estimation



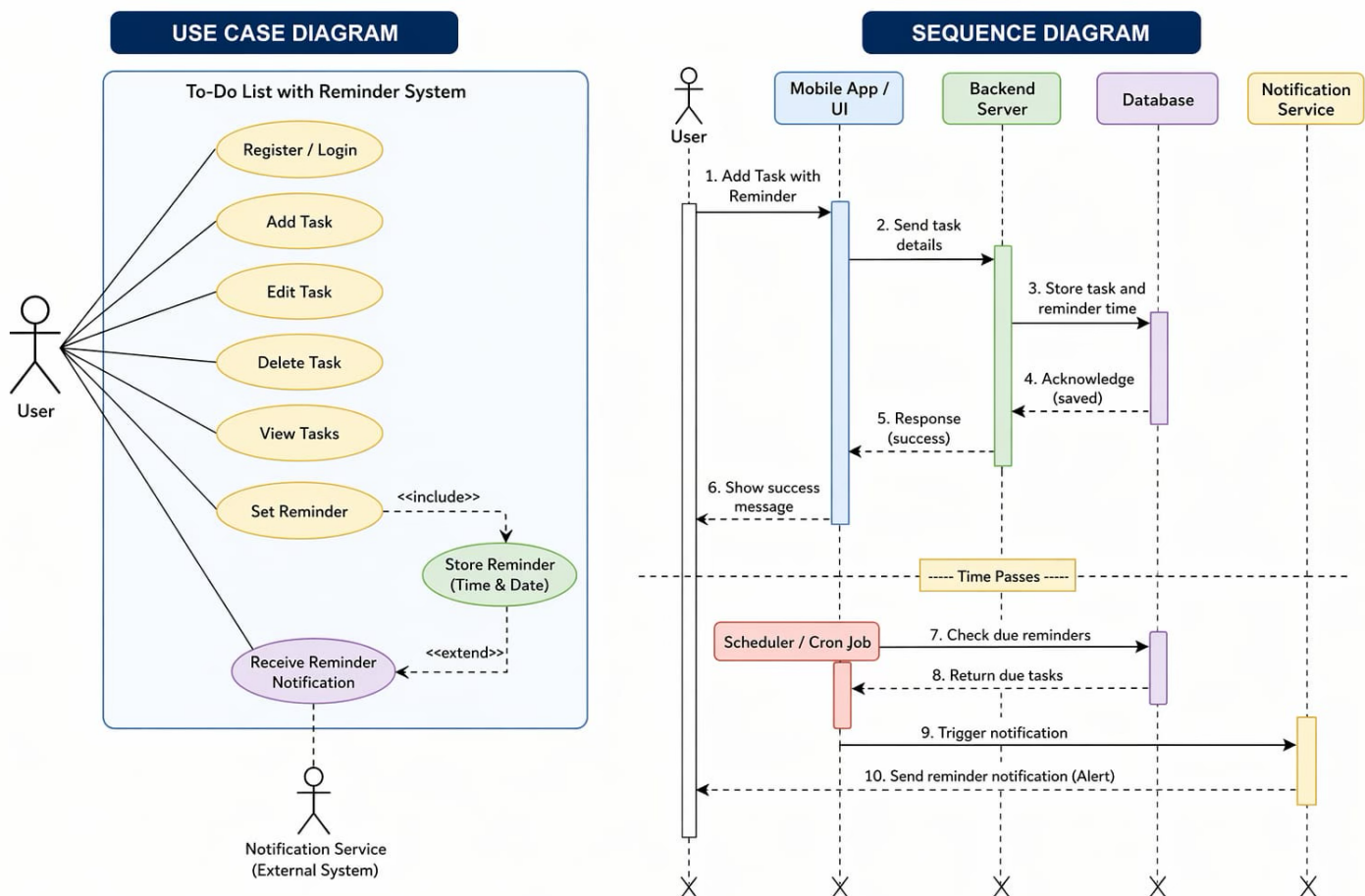
Result

The Estimation/Story Points is created for the project using Poker Estimation.

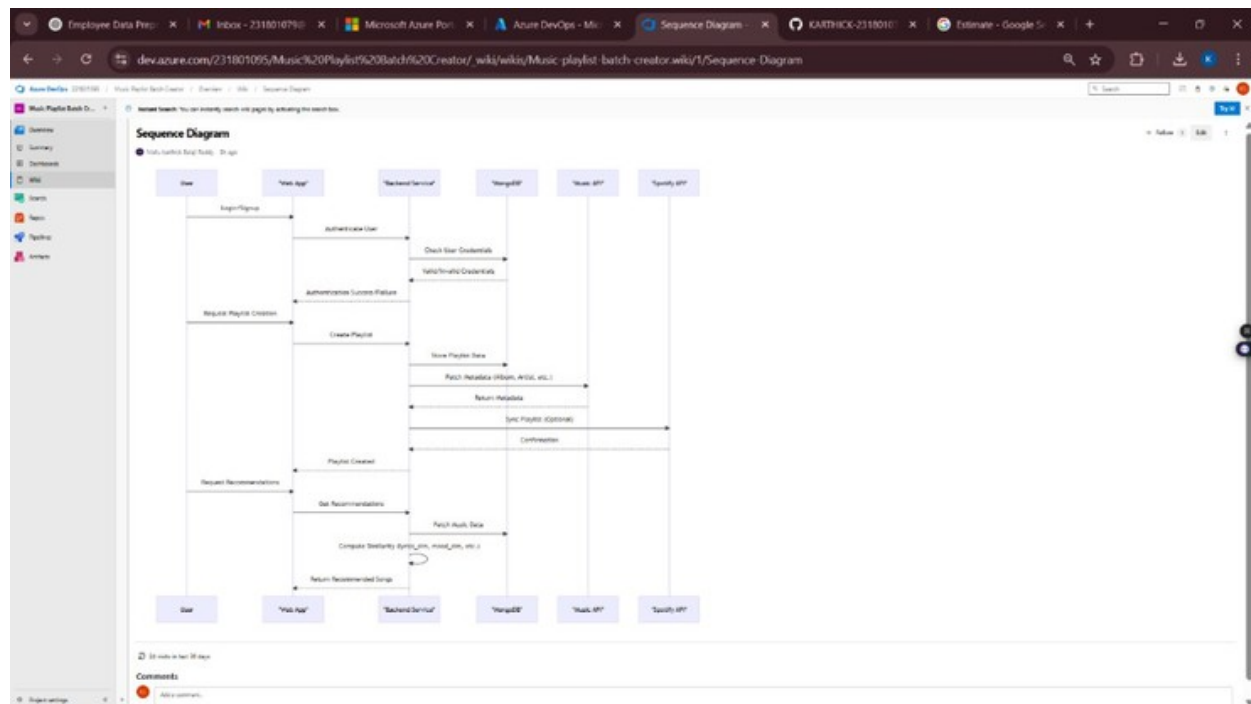
Ex.No 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

Aim: To Design a Class Diagram and Sequence Diagram for the given Project.



6A. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the todo list app Management System.

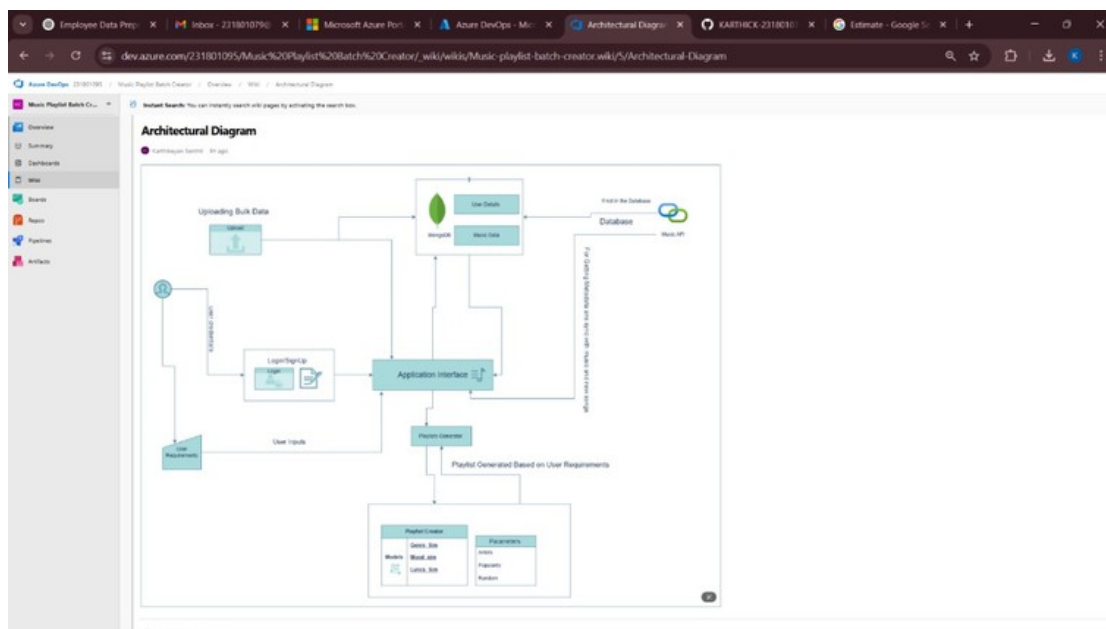
EX.NO 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

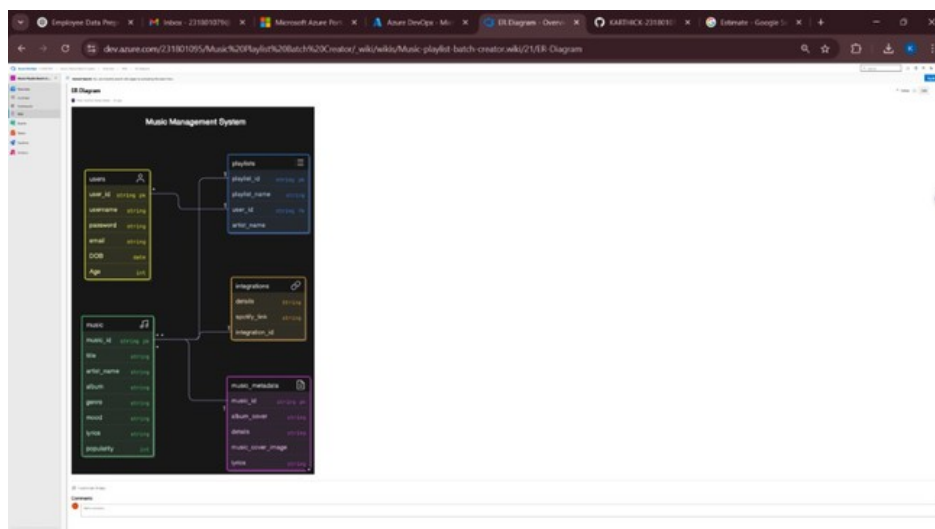
Aim:

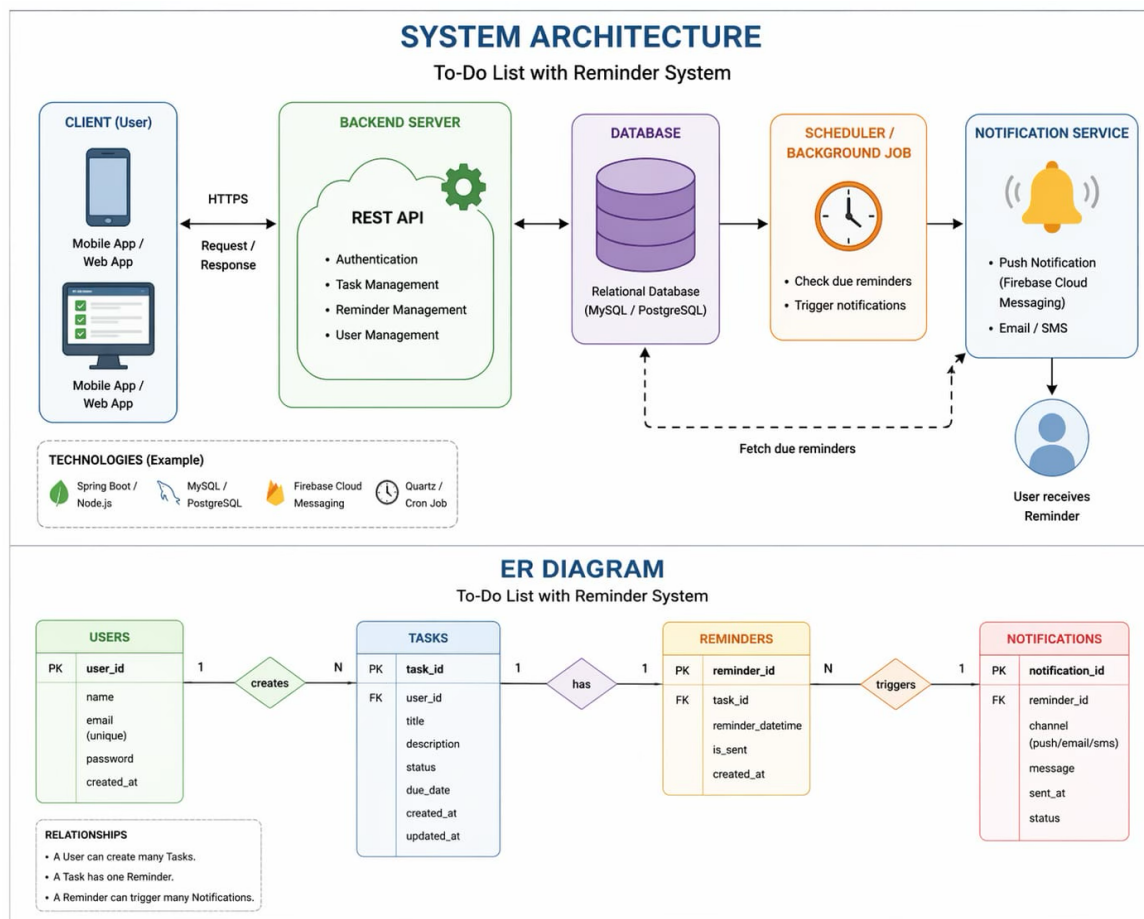
To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram





Result:

The Architecture Diagram and ER Diagram is designed Successfully for the todo list management system

EXP NO: 8

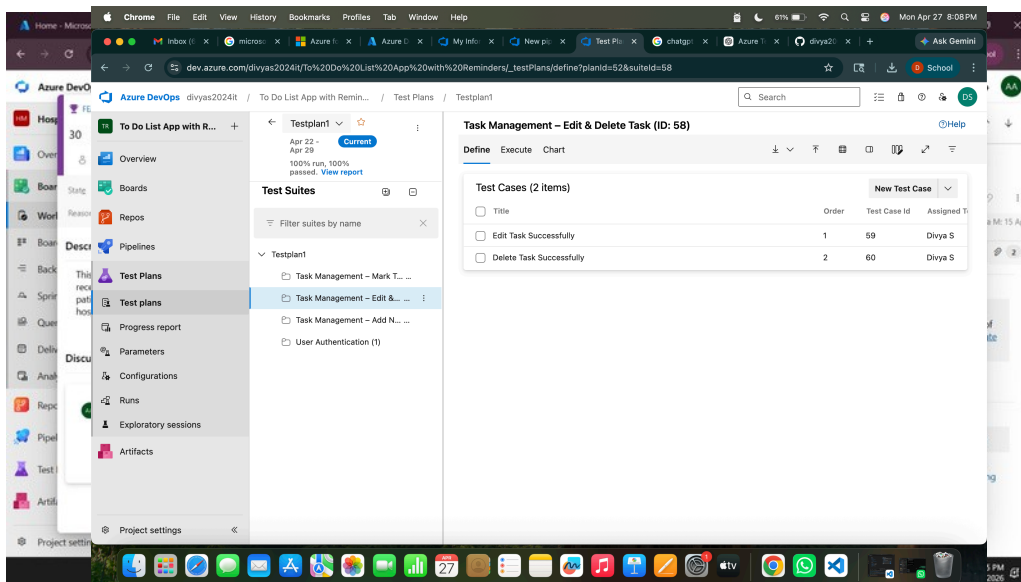
TESTING - TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

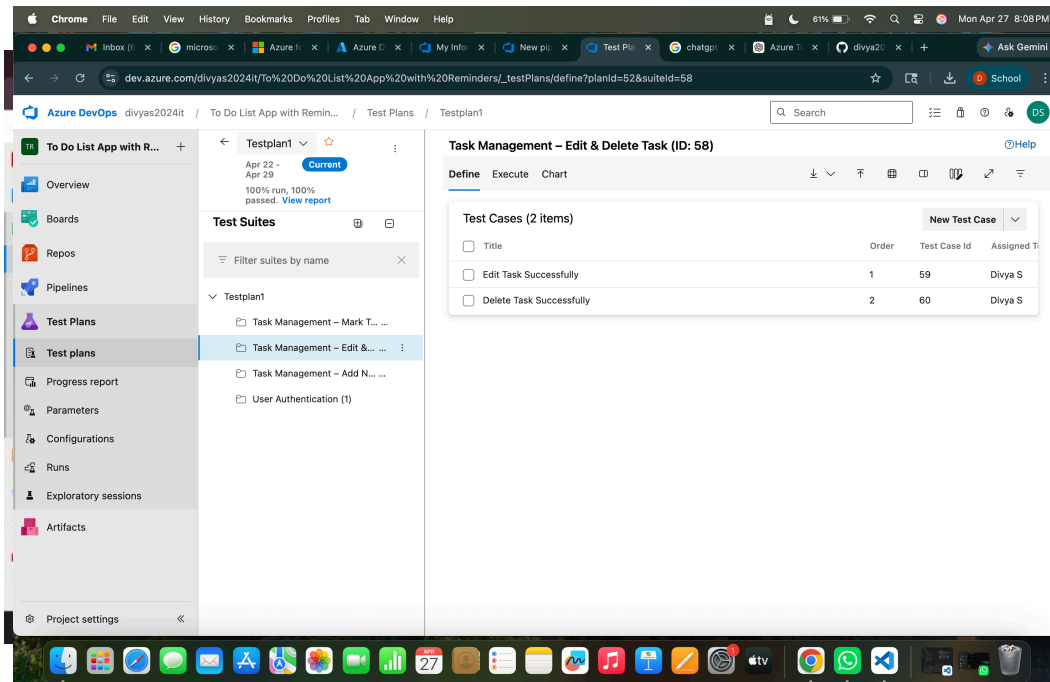
1.New test

plan



2.

3.Testsuite



Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Hospital Management System – Test Plans

USER STORIES

- As a patient, I want to register and log in securely so that I can access my medical records (ID: 101).
- As a patient, I want to book a doctor appointment so that I can schedule my consultation (ID: 102).
- As a billing officer, I want to generate invoices automatically so that billing is accurate (ID: 103). As a lab technician, I want to upload test results so that doctors can access them instantly (ID: 104).

As a doctor, I want to view my appointment schedule so that I can plan my day (ID: 105).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 - Successful Patient Registration

o Action:

• Test case

- Go to the Registration page.
- Enter valid name, email, date of birth, gender, and password.
- Click "Register".

o Expected Results:

- Registration form is displayed.
- All fields accept valid input without errors.
- Patient account is created and user is redirected to the patient dashboard.

- o **Type:** Happy Path

2. TC02 - Secure Login

- o **Action:**

- Go to the Login page.
- Enter valid email and password.
- Click on "Login".

- o **Expected Results:**

- Login form is displayed.
- Fields accept data without error.
- User is logged in and redirected to the dashboard.

- o **Type:** Happy Path

3. TC03 - Registration with Duplicate Email

- o **Action:**

- Go to the Registration page.
- Enter a name and an already registered email address.
- Click on "Register".

- o **Expected Results:**

- Fields accept data.
- Error message "Email already registered" is displayed.

- o **Type:** Error Path

4. TC04 - Login with Wrong Password

- o **Action:**

- Go to the Login page.
- Enter valid email and incorrect password.
- Click on "Login".

- o **Expected Results:**

- Input is accepted.
- Error message "Invalid username or password" is shown.

- o **Type:** Error Path

Test Suit: TS02 - Appointment Booking (ID: 111)

1. TC05 - Book Appointment Successfully

- o **Action:**

- Log in successfully.
- Search for a doctor by specialization and select an available time slot. Click "Book Appointment".

- o **Expected Results:**

- Appointment is confirmed and a confirmation notification is sent to the patient.

- o **Type:** Happy Path

2. TC06 – Book Appointment for Unavailable Slot

- o **Action:**

- Select a doctor with no available slots.
- Try to book an appointment.

- o **Expected Results:**

- Doctor has no available slots for the selected date.
- Error message "No available slots for the selected doctor" is shown.

- o **Type:** Error Path

Test Suit: TS03 - Billing & Invoice (ID: 112)

1. TC07 - Generate Invoice Successfully

- o **Action:**

- Log in as billing officer. Select a completed appointment. Click "Generate Invoice".
 - Verify the generated invoice details.
 - **Expected Results:**
 - Invoice is generated with correct service charges, patient details, and total amount.
 - **Type:** Happy Path
2. TC08 – Invoice Generation Fails Offline
- **Action:**
 - Disconnect network. Attempt to generate invoice.
 - Check the error response on the billing screen.
 - **Expected Results:**
 - Error message "Unable to generate invoice. Check your connection" is shown.
 - **Type:** Error Path

Test Suit: TS04 - Lab Test Results (ID: 113)

1. TC09 - Upload Lab Test Result Successfully
- **Action:**
 - Log in as lab technician. Select a patient test request.
 - Upload result file and click "Submit".
 - Verify the upload confirmation.
 - **Expected Results:**
 - Result is uploaded successfully and the doctor is notified.
 - **Type:** Happy Path
2. TC10 - Upload Fails with Unsupported File Format
- **Action:**
 - Attempt to upload a non-PDF/image file (e.g., .exe) as a lab result.
 - Click "Submit" without selecting a valid file.
 - Check the error message on the upload form.
 - **Expected Results:**
 - Error message "Invalid file format. Only PDF and image files are accepted" is shown.
 - **Type:** Error Path
3. TC11 - Doctor Views Appointment Schedule
- **Action:**
 - Log in as doctor. Navigate to "My Schedule".
 - View today's appointments.
 - Verify appointment details are displayed correctly.
 - **Expected Results:**
 - All scheduled appointments for the day are displayed with patient details.
 - **Type:** Happy Path
4. TC12 - Schedule Fails to Load

- o **Action:**
 - Log in as doctor. Simulate a server error.
 - Navigate to "My Schedule".
 - Check the error response on the schedule page.
 - Attempt to load the schedule.
- o **Expected Results:**
 - Error message "Unable to load schedule. Please try again later" is displayed.
- o **Type:** Error Path

Test Suit: TS05 - Pharmacy & Prescription (ID: 114)

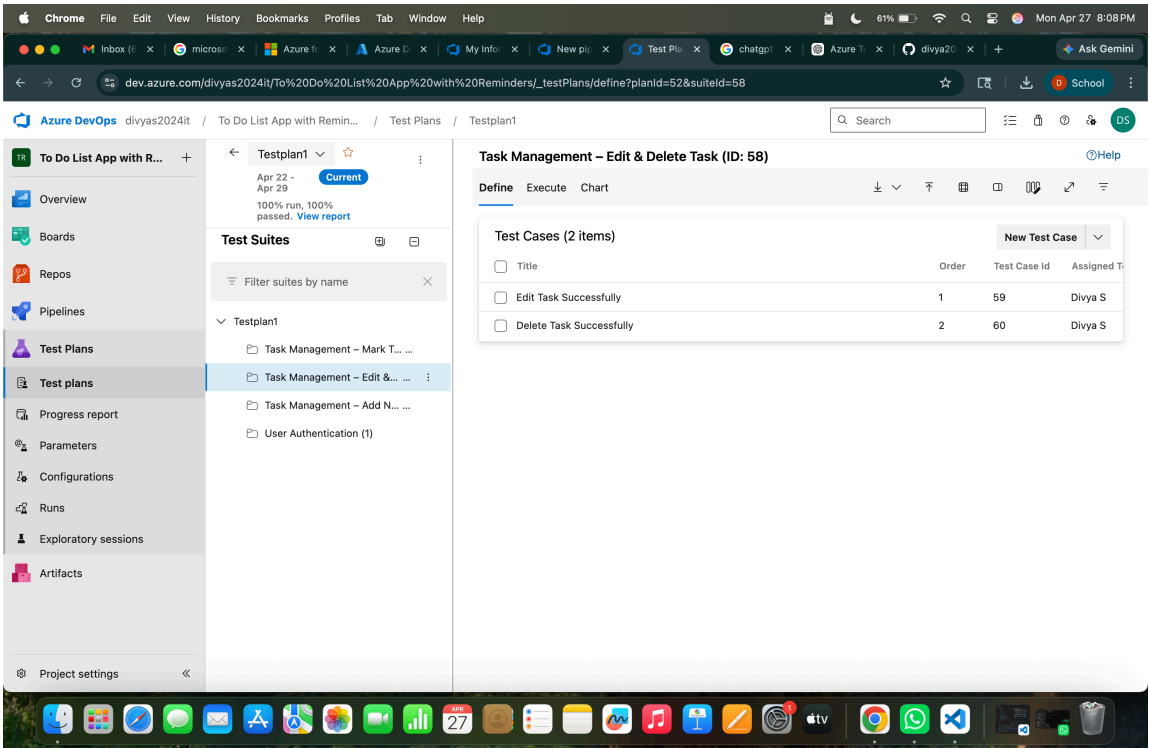
1. TC13 - Add Prescription Successfully

- o **Action:**
 - Log in as doctor. Open a patient record.
 - Click "Add Prescription". Enter medicine name, dosage, and duration.
 - Click "Save Prescription".
 - Verify the prescription is saved.
- o **Expected Results:**
 - Prescription is saved successfully and is visible in the patient's record.
- o **Type:** Happy Path

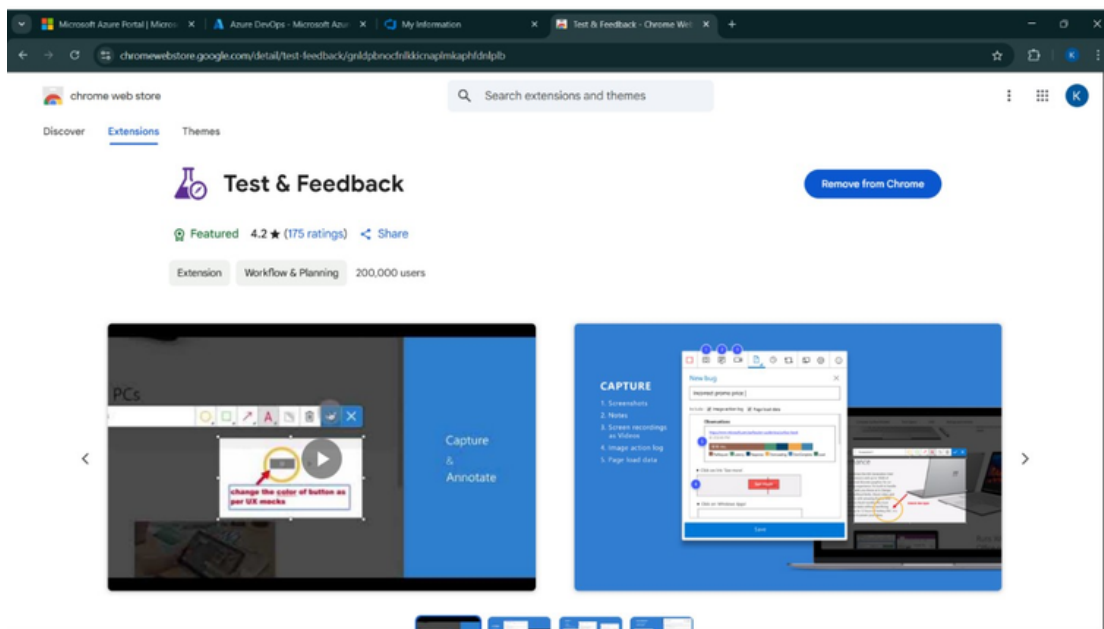
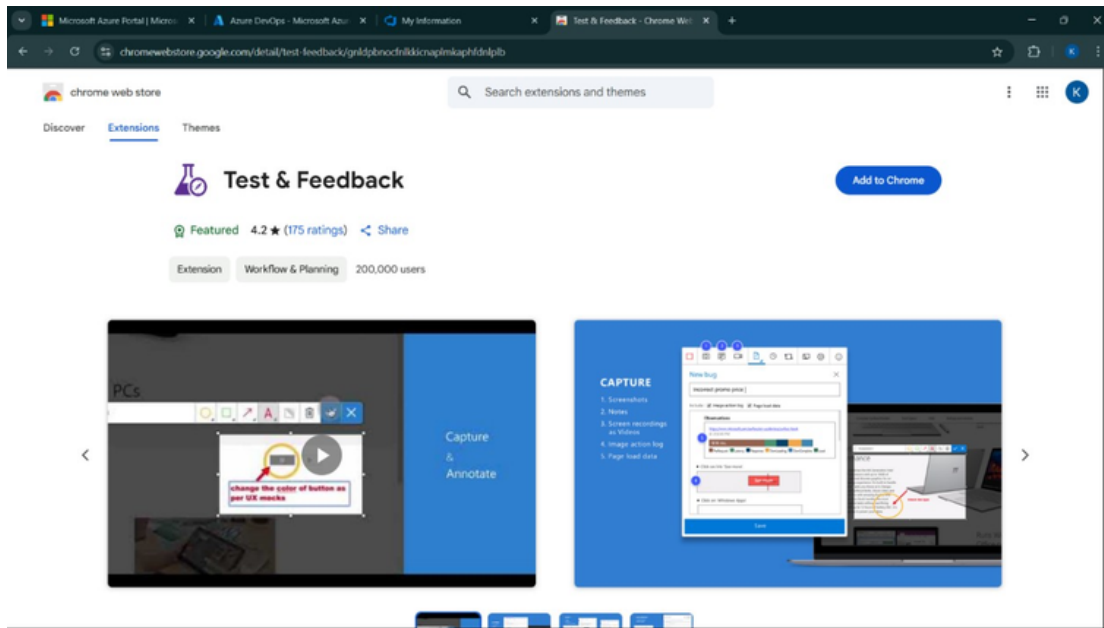
2. TC14 - Add Prescription Fails with Empty Fields

- o **Action:**
 - Log in as doctor. Open a patient record.
 - Click "Add Prescription" and leave medicine name blank.
 - Click "Save Prescription" without filling required fields.
 - Check the error response on the prescription form.
- o **Expected Results:**
 - Error message: "Please fill in all required fields" is displayed.
- o **Type:** Error Path

Test Cases

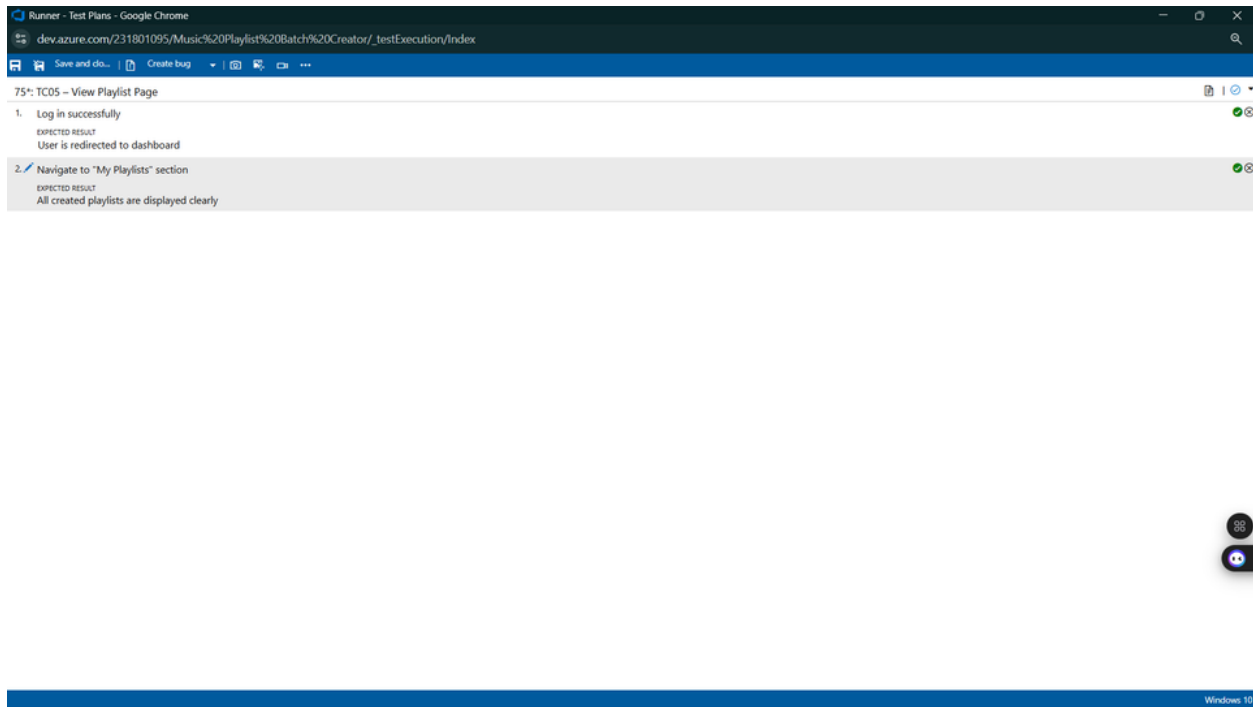


4.Installation of test

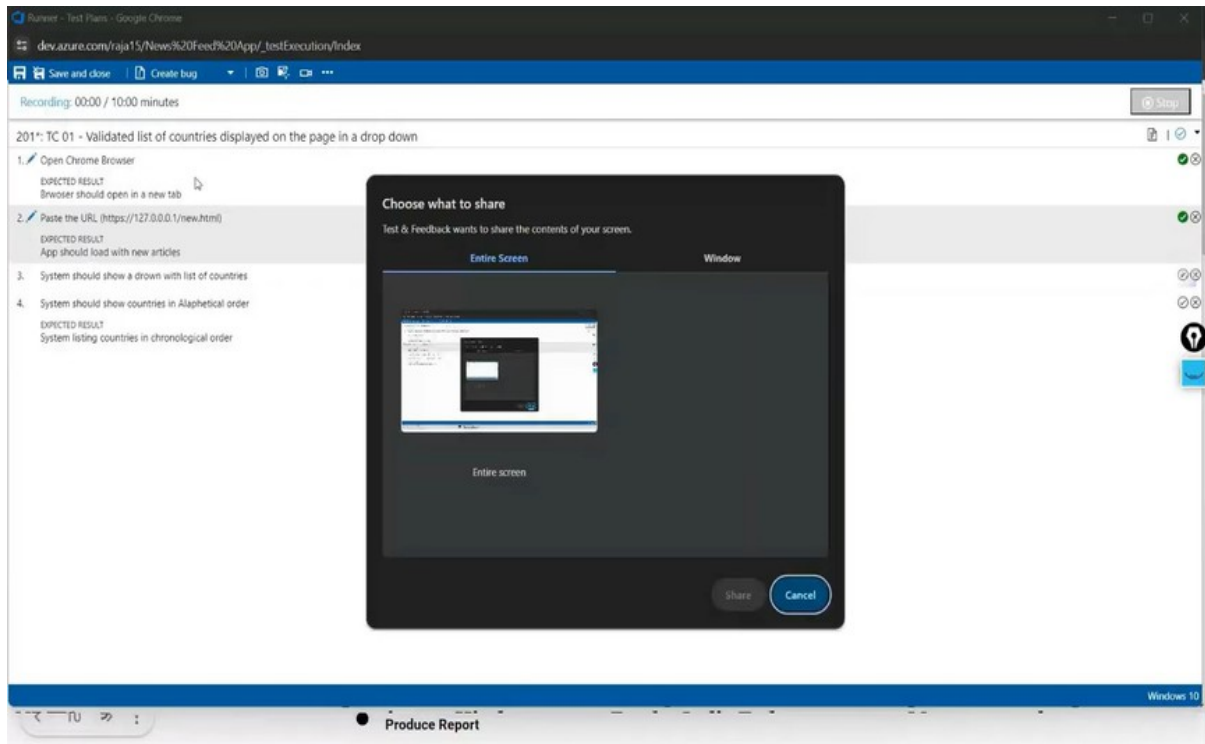


Test and feedback
Showing it as an
extension

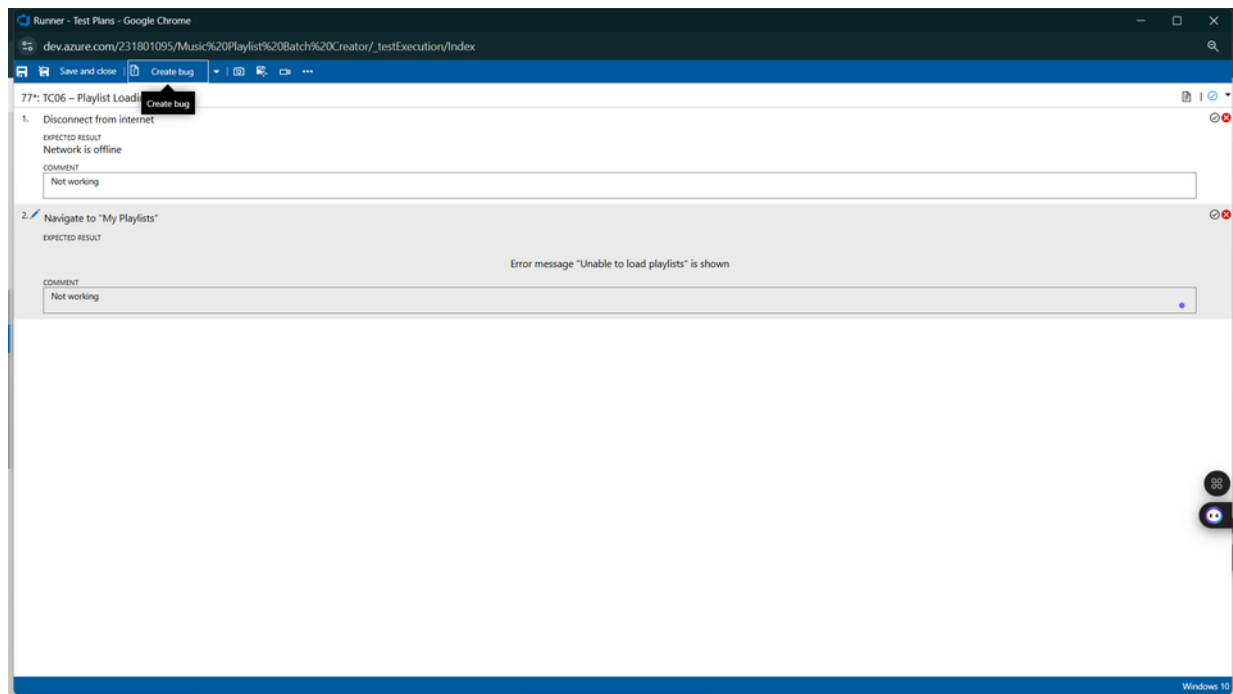
5. Running the test cases

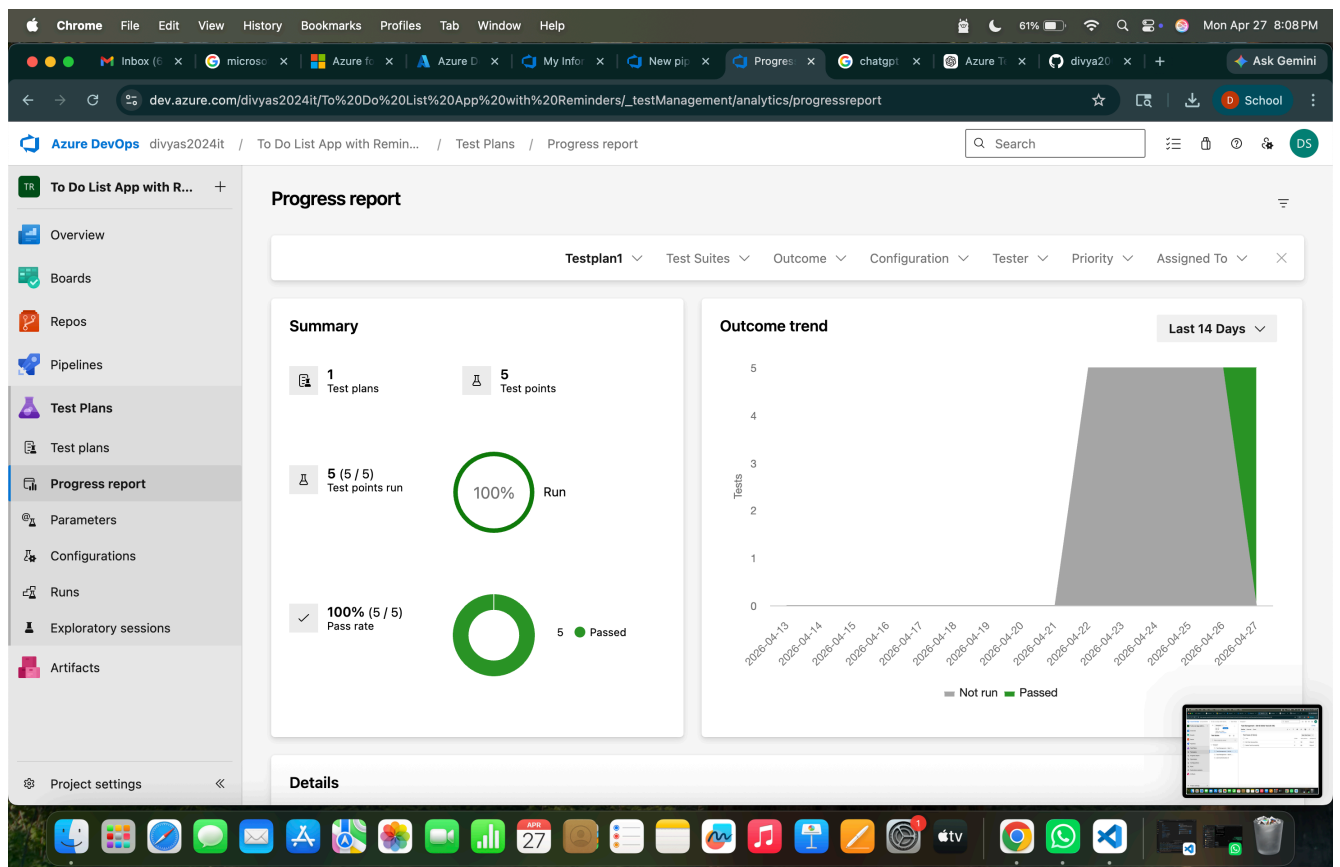


6. Recording the test case



7. Creating the bug

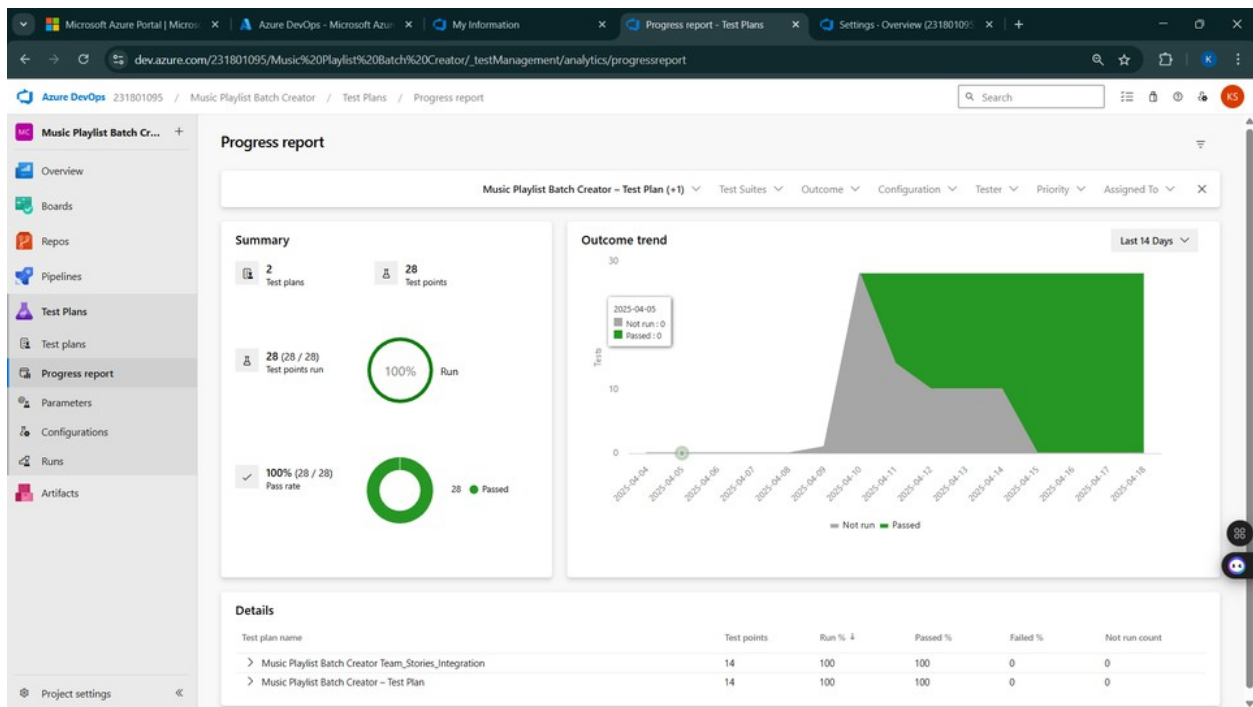




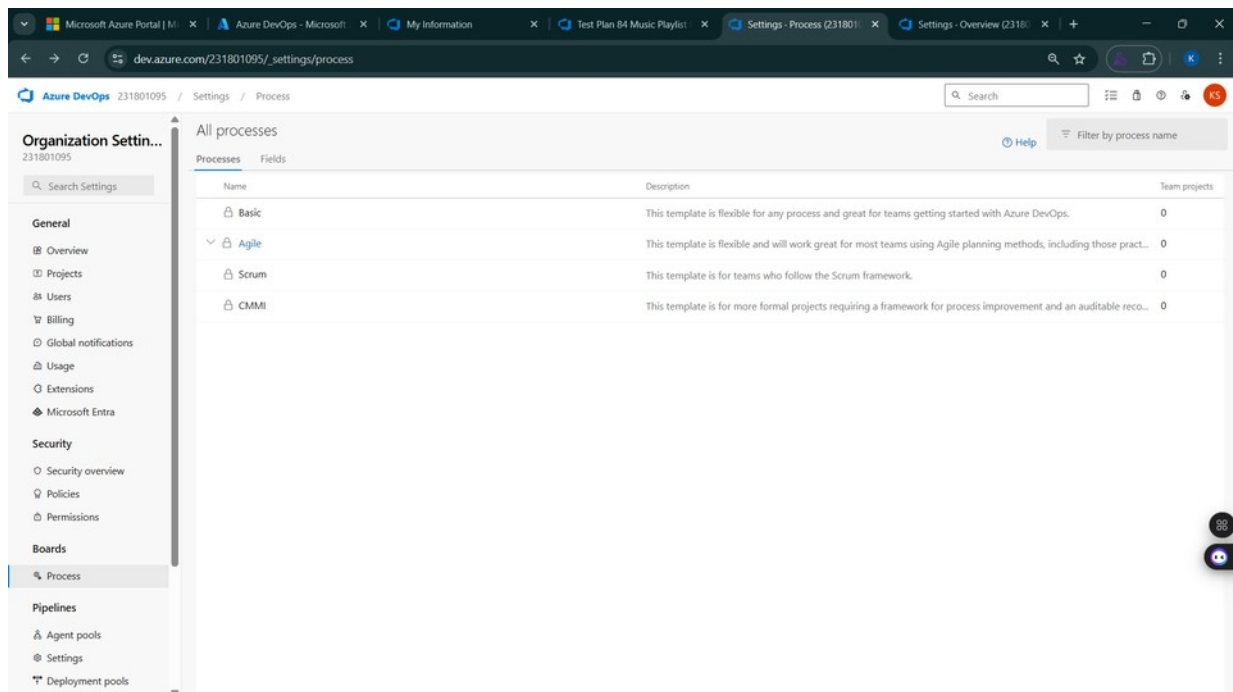
8. Test case results

The screenshot displays the Azure DevOps interface for a bug report. The bug is titled "BG 01 - Countries Drop down Not Available on the page" and is assigned to "rajesh prabhu". The bug is in the "New" state. The "Repro Step" section shows three steps: 1. Open Chrome Browser (Passed), 2. Paste the URL (Passed), and 3. System should show a dropdown with list of countries (Failed). The "Planning" section shows "Resolved Reason", "Story Points", "Priority", "Severity", and "Activity". The "Deployment" section shows "Add link" and "Related Work".

9. Test report summary



11. Changing the test template



Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan B4 Music Playlist x Settings - Process (231801095) x Settings - Overview (231801095) x + -

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan B4 Music Playlist x Settings - Process (231801095) x Settings - Overview (231801095) x + -

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

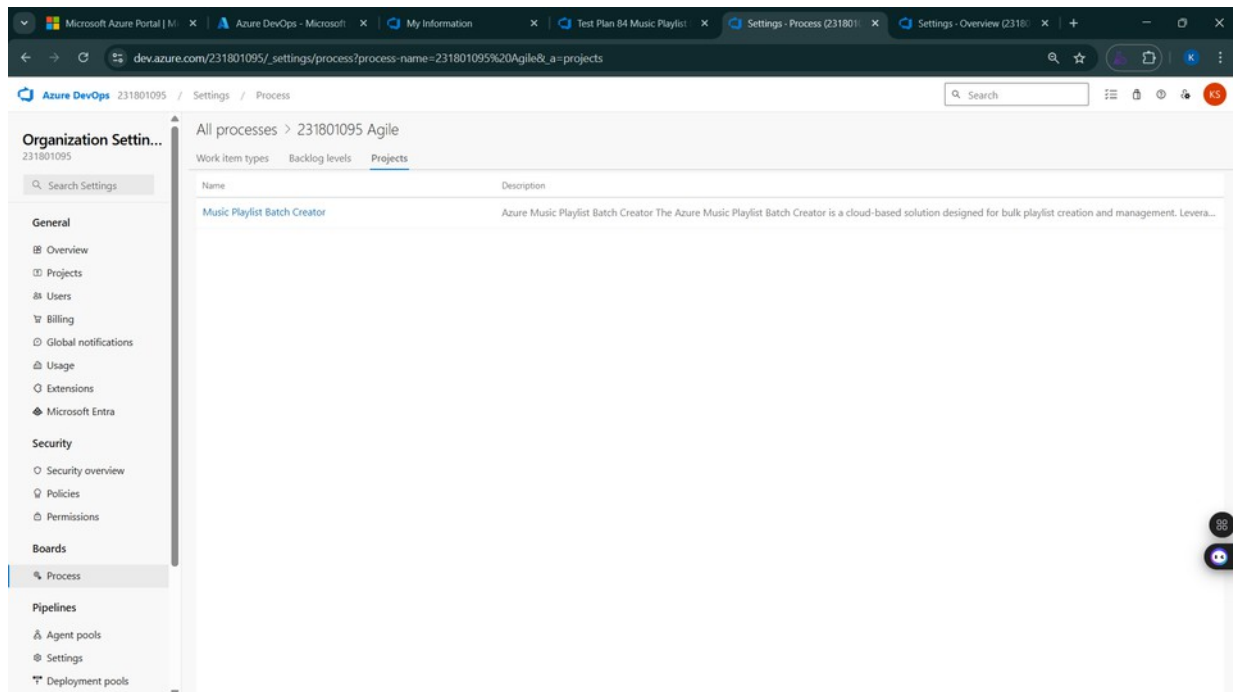
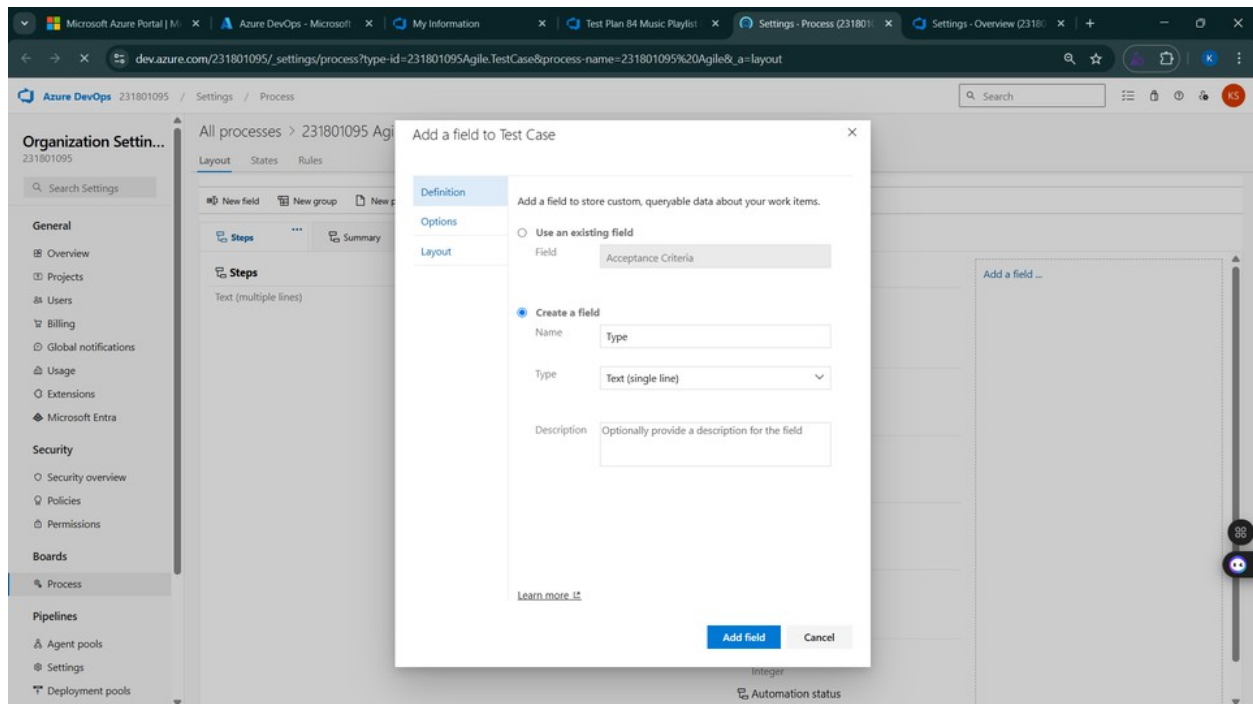
- Agent pools
- Settings
- Deployment pools

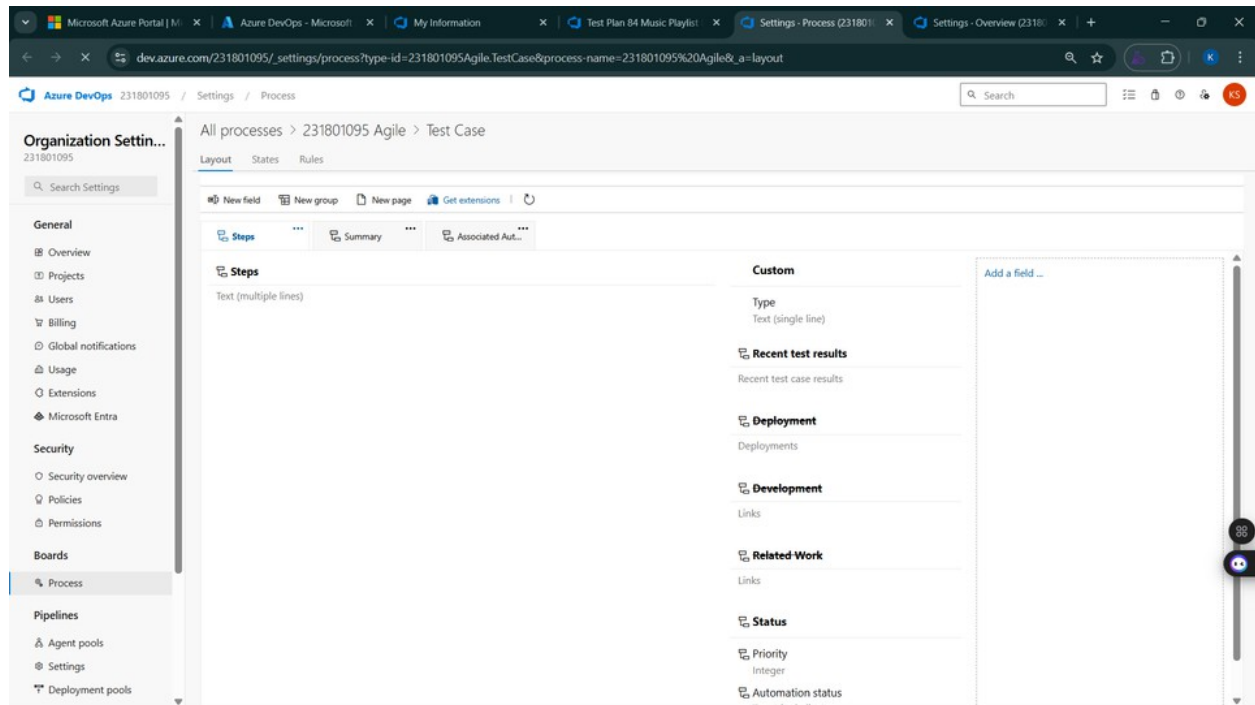
All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
231801095 Agile (default)		1
Agile Plus		0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

12. View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9	LOAD TESTING AND PIPELINES
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Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
- Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
2.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
- Click Review + Create, then Create.
5. Once deployment is complete, click Go to resource.
- 6

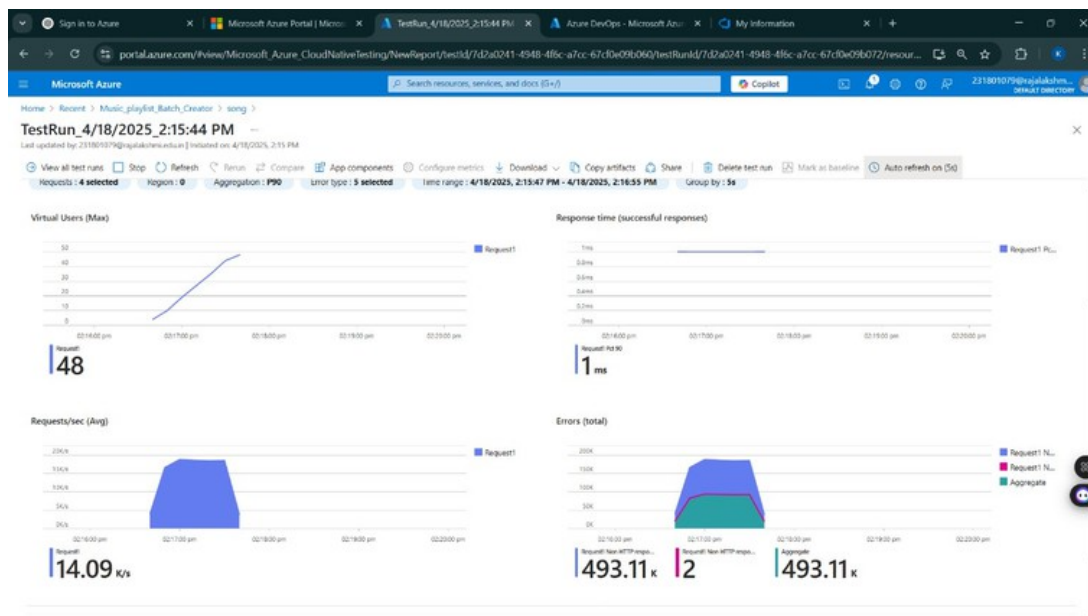
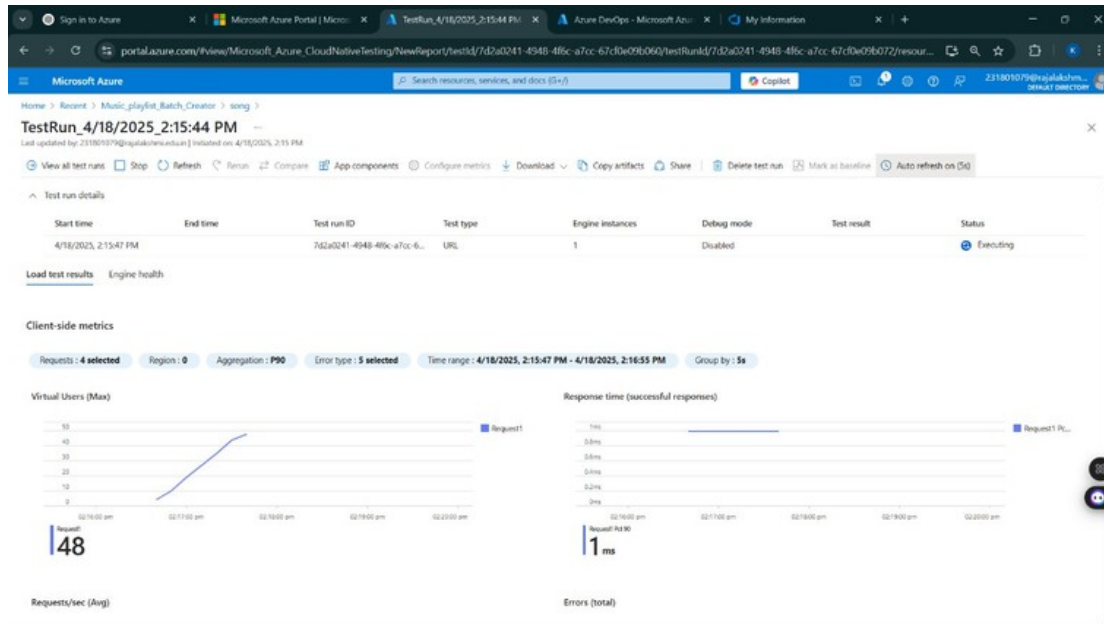
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
- Basics Tab
 2.
 - o *Test Name*: Provide a unique name.
 - - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).

- Click Review + Create → Create to start the test.

Load Testing



Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Node.js/Express-based Hospital Management System project with Azure DevOps. The pipeline will automatically install dependencies, run tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.
 - o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for Node.js and Express:

yml Code

trigger:

- main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

Step 1: Checkout the code from GitHub

- checkout: self

Step 2: Set up Node.js environment

- task:

NodeTool@0

inputs:

versionSpec: '18.x' # Use the latest Node.js 18.x version

displayName: "Set up Python"

Step 3: Install dependencies from the correct path

- script: |

python -m pip install --upgrade pip

npm install # Install HMS project dependencies

displayName: "Install dependencies"

Step 4: Run a simple Python script to check the environment

- script: |

node -e "console.log(' Hospital Management System pipeline running successfully!')" displayName: "Run a Python script"

3. Pipeline Tasks Include:

- o Setting up the Node.js environment using the NodeTool task.
- o Installing project dependencies using npm install from package.json.
- o Running unit tests using npm test to verify all HMS modules work correctly.

4.Run and Monitor Pipeline:

- o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- o Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps web interface for a project named "Music Playlist Batch Creator". The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines, Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main content area shows the details for a specific pipeline run, "#20250424.3 • Pipeline 2", which was manually triggered by user "Karthick S". The run status is "Success" and it completed in 65 seconds. The interface also shows a "Jobs" table with one job named "Job" that was successful. A message at the top indicates that the run is being retained as one of the 3 most recent runs by the main branch.

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_build/results?buildId=308&view=results

Azure DevOps 231801095 / Music Playlist Batch Creator / Pipelines / Music Playlist Batch Creator... / 20250424.3

#20250424.3 • Pipeline 2
Music Playlist Batch Creator (9)

This run is being retained as one of 3 recent runs by main (Branch). [View retention leases](#)

Summary Code Coverage

Manually run by **Karthick S** [View 52 changes](#)

Repository and version: Music Playlist Batch Creator, main, a87bd670

Time started and elapsed: Just now, 24s

Related: 0 work items, 0 artifacts

Tests and coverage: [Get started](#)

Jobs

Name	Status	Duration
Job	Success	65

Project settings

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

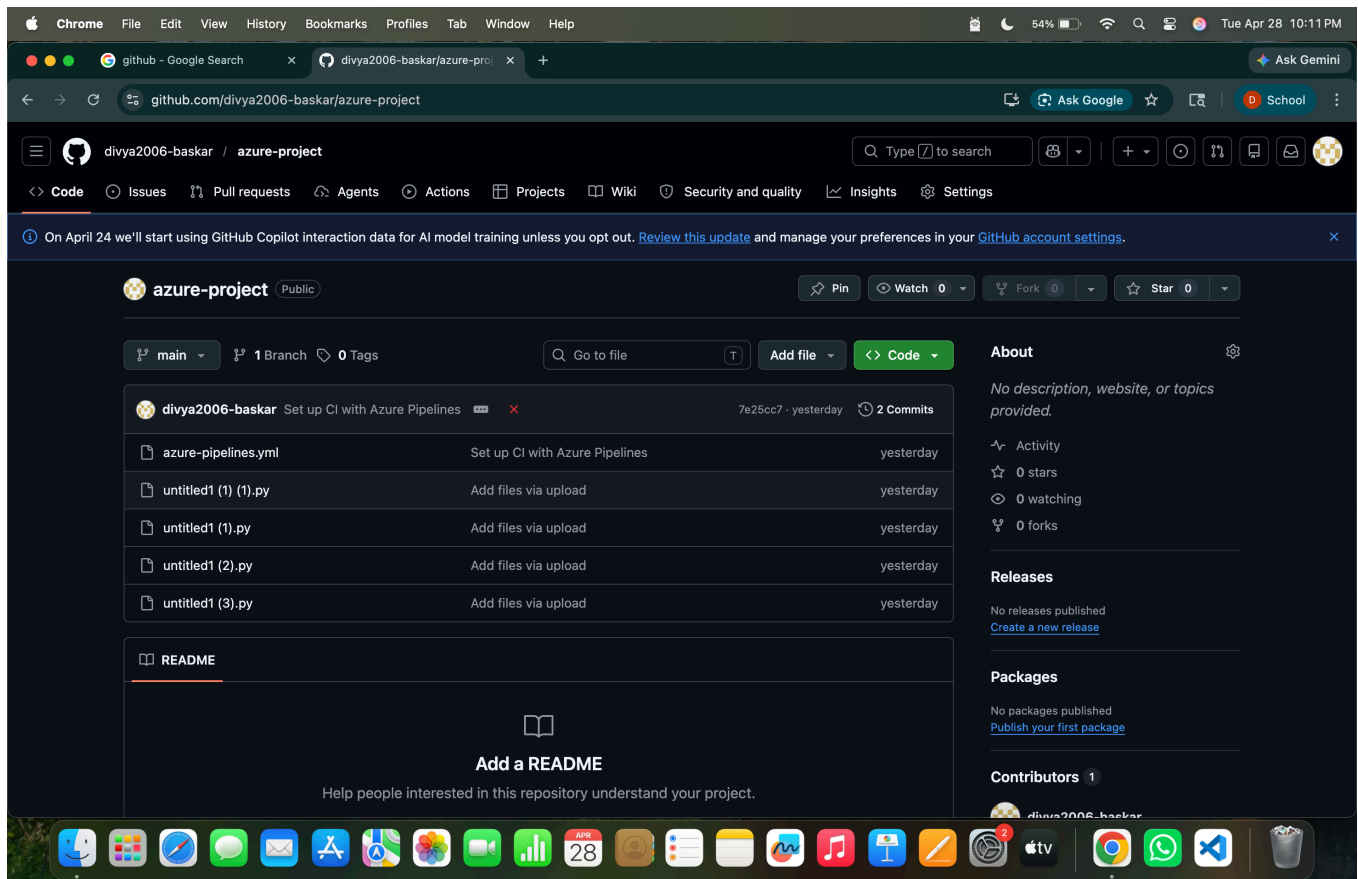
EXP NO: 10

GITHUB:PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the Todo List Management System folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the HMS project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.